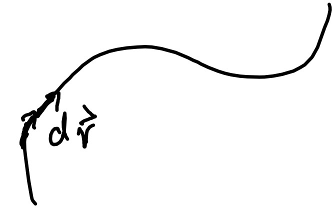
Recall $\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\,dt$



also write $\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \vec{T}\,ds$ where $\vec{T}$ is unit tangent as usual

or $\int_C \vec{F} \cdot d\vec{r} = \int_C P\,dx + Q\,dy + R\,dz$ where $\vec{F}(x,y,z) = (P(x,y,z), Q(x,y,z), R(x,y,z))$

$= \int_C P\,dx + Q\,dy$ where $\vec{F}(x,y) = (P(x,y), Q(x,y))$

Evaluation is the same! Don't magically get to forget the curve

$$\int_C P\,dx + Q\,dy$$

In fact, we'd determine $\vec{r}(t) = (r_1(t), r_2(t))$, write $r_1(t) = x(t)$, $r_2(t) = y(t)$, then

$$\int_a^b P(x(t), y(t))\frac{dx}{dt}\,dt + Q(x(t), y(t))\frac{dy}{dt}\,dt$$

notice $\frac{dx}{dt} = r_1'(t)$ $\frac{dy}{dt} = r_2'(t)$

$$= \int_a^b \big(P(x(t), y(t)), Q(x(t), y(t))\big)\cdot\left(\frac{dx}{dt}, \frac{dy}{dt}\right)dt$$

$$= \int_a^b \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt$$

Same thing all the way! Just alternative notation.

Thm: Let $\vec{F}: D \subset \mathbb{R}^2 \to \mathbb{R}^2$ with $\vec{F} = (P, Q)$. If $\vec{F}$ is conservative, and the functions $P, Q : D \to \mathbb{R}$ have continuous first order partial derivs, then

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

Proof: $\vec{F} = (f_x, f_y)$ for some $f : D \subset \mathbb{R}^2 \to \mathbb{R}$.

$f_x = P \Rightarrow P_y = f_{yx}$

$f_y = Q \Rightarrow Q_x = f_{xy}$

and $P_y = f_{yx}$ and $Q_x = f_{xy}$ are continuous so

$f_{xy} = f_{yx} \Leftrightarrow \dfrac{\partial Q}{\partial x} = \dfrac{\partial P}{\partial y}$

by Clairaut-Schwartz

We now want to relate the line integral $\oint_C \vec{F}\cdot d\vec{r}$ of a 2D v.f. $\vec{F}=(P,Q)$ along a <u>simple</u> <u>closed</u> curve C and a double integral $\iint_D g(x,y)\,dx\,dy$ where D is the interior domain to the curve C, and $g(x,y)$ is a suitable f^n obtained by some partial differentiation of P and Q.

Note: we also require that C is positively oriented, i.e. counterclockwise.

Thm: Green's Theorem

Let C be a piecewise regular simple closed curve with counterclockwise orientation. Let D be the interior domain to the curve C. Let U be some subset of $\mathbb{R}^2$ which contains D and C and let $\vec{F} = (P, Q) : U \to \mathbb{R}^2$ be a 2D vector field whose components are of class C^1. Then

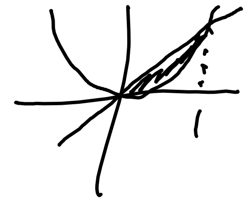
$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dx\,dy$$

easy remember: $\overline{\partial x} - \overline{\partial y} \; dx\,dy$

Side note: what can you say when $\vec{F}$ is conservative?

e.g. let's confirm this for an example.

D domain enclosed by $y = x$ and $y = x^2$



let $\vec{F}(x,y) = (x^2y, 2xy) \ \forall (x,y) \in \mathbb{R}^2$.

$P = x^2y \qquad Q = 2xy$

$$\iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dx\,dy = \iint_D 2y - x^2\, dx\,dy = \int_0^1 \int_{x^2}^{x} 2y - x^2\, dy\, dx$$

$$= \int_0^1 \left[y^2 - yx^2 \right]_{x^2}^{x} dx = \int_0^1 x^2 - x^3 - (x^4 - x^4)\, dx = \int_0^1 x^2 - x^3\, dx$$

$$= \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

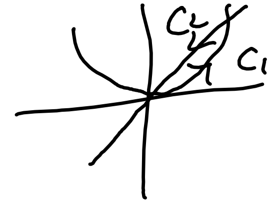
$C_1 : (t, t^2)$ $t: 0 \to 1$

$\quad\;\; x(t) \;\; y(t)$

$C_2\ (t, t)$ $t: 1 \to 0$

$\quad\;\; x \;\; y$

$F = (x^2 y, 2xy)$



$$\int_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} =$$

$$\int_0^1 (t^2 t^2, 2t t^2) \cdot (1, 2t)\, dt + \int_1^0 (t^2 t, 2tt) \cdot (1,1)\, dt$$

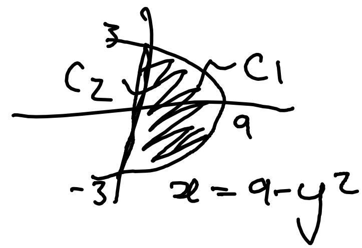
$$= \int_0^1 t^4 + 4t^4\, dt + \int_1^0 t^3 + 2t^2\, dt = \int_0^1 \quad + -\int_0^1$$

$$= \int_0^1 t^4 + 4t^4 - t^3 - 2t^2\, dt = \int_0^1 5t^4 - t^3 - 2t^2\, dt$$

$$= t^5 - \frac{t^4}{4} - \frac{2t^3}{3} \Big]_0^1 = 1 - \frac{1}{4} - \frac{2}{3} = \frac{12 - 3 - 8}{12} = \frac{1}{12}$$

yay!

e.g. $F = \left(\frac{y^2}{2}, \frac{x^2}{4} \right)$



$C_1: \; (\underset{x(t)}{9-t^2}, \underset{y(t)}{t}) \qquad t: -3 \to 3$

$C_2: \; (\underset{x(t)}{0}, \underset{y(t)}{t}) \qquad t: 3 \to -3$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} + \int_{C_2} = \int_{-3}^{3} \left(\frac{t^2}{2}, \frac{(9-t^2)^2}{4} \right) \cdot (-2t, 1)\, dt$$

$$+ \int_{3}^{-3} \left(\frac{t^2}{2}, \frac{0}{4} \right) \cdot (0, 1)\, dt = \int_{-3}^{3} -t^3 + \frac{1}{4}(9-t^2)^2 \, dt$$

$$+ \int_{3}^{-3} 0 \, dt = \int_{-3}^{3} -t^3 + \frac{1}{4}(81 - 18t^2 + t^4)\, dt \qquad \text{eval!}$$

$$= \frac{324}{5}$$ (integral-calculator.com)

or: $$\iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dx\,dy = \iint_D \frac{1}{2}x - y\, dx\,dy$$

$$= \int_{-3}^{3}\int_{0}^{9-y^2} \frac{1}{2}x - y\, dx\,dy = \int_{-3}^{3} \left.\frac{x^2}{4} - xy\right]_0^{9-y^2} dy =$$

$$= \int_{-3}^{3} \frac{1}{4}(9-y^2)^2 - (9-y^2)y\, dy = \int_{-3}^{3} \frac{(81 - 18y^2 + y^4)}{4} - 9y + y^3\, dy$$

$$= \frac{324}{5}$$

Note: you can do the $\oint_C$ part without t, just work in x. Same thing at the end of the day. Check notes!

I find t less confusing for curves, keeps things distinct.